Broadband transport — the synchronous digital hierarchy

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All data networks require a physical transmission medium to convey information between network nodes. Within a local environment this physical medium might, for example, take the form of an Ethernet LAN, but wide area connections are provided using telecommunications constant bit rate transmission equipment. Furthermore, the assumption that data networking is simply the provision of WAN connectivity for large corporate networks is becoming dated. The explosion of interest in the Internet means that, for transport networks, the term data may encompass voice, video and multimedia applications for delivery to both home and office. This places additional requirements on the network infrastructure as each service has specific transport requirements.

Network operators are currently in the process of deploying core networks of equipment conforming to the ITU-T Recommendations for a synchronous digital hierarchy (SDH), but many networks also contain a significant proportion of older transmission technologies. This paper provides a review of transmission technology and describes the impact of such networks on the transport of data.

1. Introduction

The ITU have chosen equipment conforming to the synchronous digital hierarchy (SDH) as the transport layer for B-ISDN networks [1]. However, for many operators the majority of their network consists of the older plesiochronous digital hierarchy (PDH) equipment. This paper describes the development of modern transport networks, the origins and advantages of the synchronous digital hierarchy and the application of the technology to the transportation of data services. The intention is to provide a description of the behaviour and characteristics of modern transport networks so that designers of client networks are able to assess the implications for a specific service. It should be remembered that such transport networks underpin every service offered by modern communications companies. In this sense, data must be viewed in its widest interpretation to include voice, video and other services. Furthermore, this paper provides a review of transport technology for the core network but does not consider access technologies such as xDSL and PONs.

The standards for the synchronous digital hierarchy were developed from those which originated in North America for a synchronous optical network (SONET) [2]. The two hierarchies share common defining principles and indeed are compatible at specific bit rates, but have developed in isolation and hence serve different masters. Specific differences between SONET and SDH interfaces may result in practical difficulties in interoperating equipment. Some of these difficulties are described below.

2. The layered transport network

Modern telecommunications networks are required to transport many different services. The days when operators would deploy specific network technologies to support specific services are surely numbered. The reasons for this are clear — the costs of providing a network are enormous and in order to maximise the return on that investment, the revenue needs to be generated from more than one service.

The commercial pressures to maximise the benefit of each technology have led to the development of a layered architecture for networks, such as that illustrated in Fig 1.

The concept of network layering has found wide acceptance internationally and has been used as the basis of both network architecture and equipment functionality recommendations [3, 4].

Modern transport networks can be partitioned into networking layers and the physical transmission layers. Both affect the overall transport of data services, but it is the latter which are the subject of this paper.

Figure 1 illustrates some of the layers which may be present in a network using SDH technology. Network layering brings some disadvantages in that it is not always obvious in which layer a certain functionality should reside. For example, an IP service may be carried over an

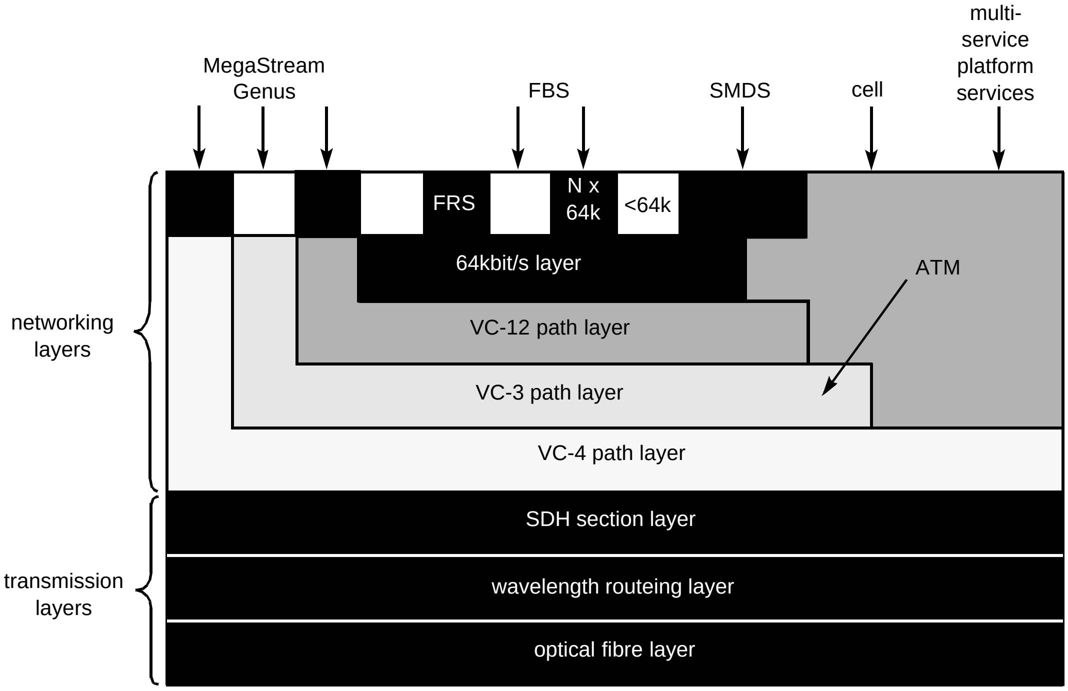

Fig 1 Network layers.

asynchronous transfer mode (ATM) network using SDH transmission. In this case resilience may reside in each layer and there is a need to consider the entire service in order to avoid contention between protection mechanisms and wasted bandwidth.

The important principle behind network layering is that each layer may be managed independently, and may be subject to evolution or replacement without affecting other layers provided that the transfer characteristics and inter-layer interfaces are maintained.

The interfaces to the transport layers of such a network were determined by historical development which is described in the following section.

3. The plesiochronous digital hierarchy

The transmission networks of many operators have evolved over a number of years to suit changing requirements. Although overlay networks have been added to meet the needs of specific new services, most networks have grown to serve the interests of the dominant source of traffic over the period — the plain old telephony service (POTS) — and usually consist of fixed transmission links between switching centres.

The size of the switching centre, and the capacity of the links between them, have been determined by static measurements of traffic flow. The resultant network therefore consisted of a network of switching nodes interconnected by a number of high-capacity (140 and 565 Mbit/s) pipes as illustrated in Fig 2.

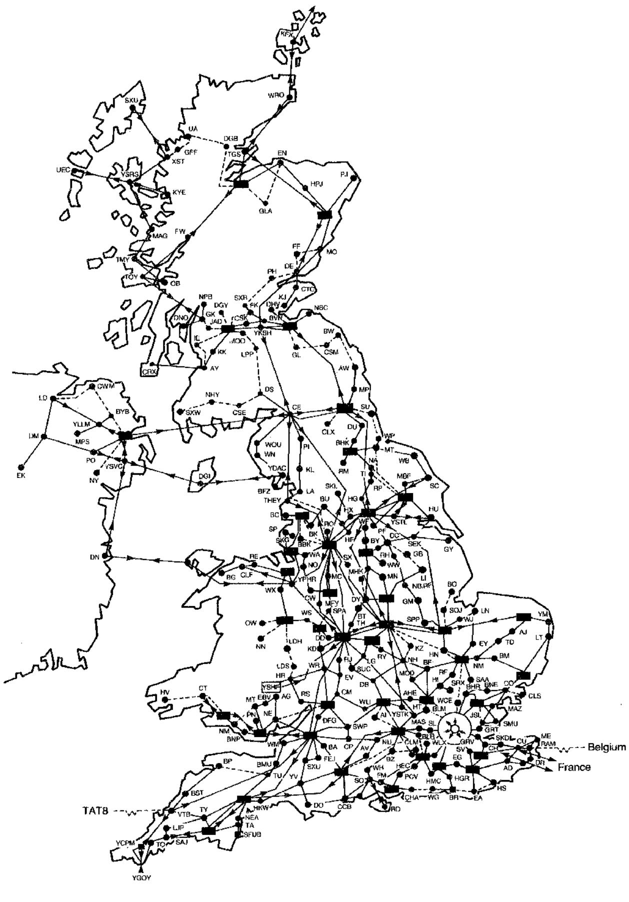

Fig 2 BT PDH transmission network.

Digital transmission is used between switching centres and, in order to maximise the efficiency of the optical line transmission system (OLTS), the data between sites is concentrated before transmission. In a typical network, each link between 64 kbit/s switches may involve several stages of multiplexing, then optical or electrical transmission, followed by several stages of demultiplexing. This has resulted in a 'multiplex mountain' as illustrated in Fig 3.

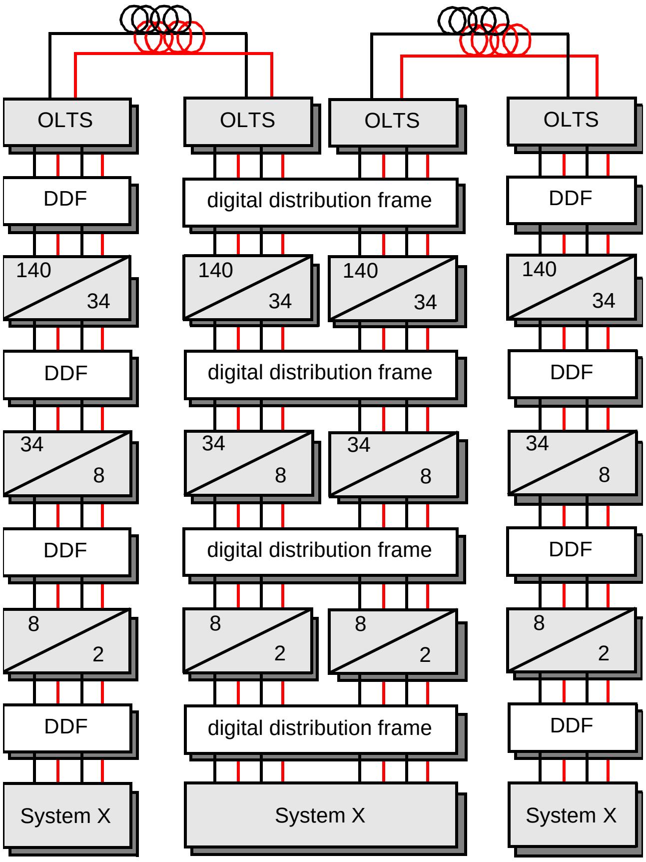

Fig 3 PDH 'multiplex mountain'.

The plesiochronous digital hierarchy, as employed in Europe, is based on 30-channel pulse code modulation systems for speech. The signals are multiplexed through a hierarchy of 8, 34, 140 and 565 Mbit/s before transmission [5]. All of the cables connecting switches to multiplexers and transmission systems are hard wired on a digital distribution frame (DDF). The interfaces at these rates are well established and defined internationally [6] which has enabled their adoption for a wide variety of data CPE.

However, the existing PDH transmission networks exhibit a number of disadvantages:

- 'dumb' equipment with limited operation and maintenance information,
- limited remote management capability,
- poor reliability,
- manual circuit (re-)configuration,
- only electrical interfaces are fully specified,
- the hierarchy was only specified up to 140 Mbit/s, and the hierarchies are different in Europe, and the USA and Japan.

The synchronous digital hierarchy was developed in order to address these perceived deficiencies.

PDH transmission systems accommodate frequency differences between different payloads through a process known as bit justification. This means that the aggregate transmission rate is always greater than the sum of the client payloads being transported; the difference being accommodated by the introduction of 'stuffing' bits. The process introduces phase variations in the payload being transported which are known as jitter.

Also, the use of a multiplex hierarchy introduces a large volume of electronics into the path which degrades the reliability.

The vast majority of the BT digital network employs optical transmission. The underlying or background error rate on such systems during their operational lifetime may typically be very low, with less than one error in 1×10^{11} bits. However, the actual error performance of digital paths may be somewhat worse than this due to the effects of system ageing and electrical interference. The latter effect is thought to be the most dominant short-term error mechanism in existing networks. The dominant influence on the availability of a circuit is probably that of cable damage.

4. Data platform physical interfaces

The BT digital transmission network was originally developed to serve the requirements of PSTN traffic which is based on 30 speech channels within a 2048 kbit/s digital signal. It is uneconomic to transport these signals over optical fibre and so several channels are multiplexed together before transmission. This has led to the definition of a number of different hierarchical bit rates which are used for data interfaces.

Typically, data customer premises equipment may include interfaces at 2 Mbit/s (or lower), 8 Mbit/s, 34 Mbit/s and 140 Mbit/s. To such CPE, the PDH transmission network essentially constitutes a transparent technology. Transported information may be degraded by transmission (errors and jitter may be introduced), but PDH systems are just transmission pipes! This cannot be said of SDH equipment — it cannot be considered to be 'managed PDH'.

SDH interfaces are now becoming more common on ATM switches, IP routers, etc, but in truth the majority of these are SONET rather than true SDH interfaces. The impact of this distinction may be subtle differences in the implementation of specific bits within a single byte which result in the generation of spurious alarms. However, more fundamental differences exist such as the use of incompatible types of fibre and connector. The impact on the network operator is a need to be aware of the differences and to realise that SDH interfaces on CPE may not be ‘plug and play’.

5. The synchronous digital hierarchy

The synchronous digital hierarchy is essentially a new multiplexing structure for digital transmission systems. It was developed to address a number of problems with the existing digital transmission network.

The synchronous digital hierarchy was designed to follow the principles of network layering which are fully defined in ITU-T Recommendation G.803 [3] and illustrated in Fig 1. According to these principles, the transport layer may be divided into a number of independent layers. Each layer acts as a server for those layers above it in the hierarchy and as a client for the layers below. Information is adapted for transmission by an adaptation function within the server layer and a layer-specific overhead added by a trail termination function within the server layer in order to monitor the client data through the layer (and any lower layers). This is illustrated in Fig 4, where the notation used is consistent with that employed in ITU-T Recommendation G.803 [3].

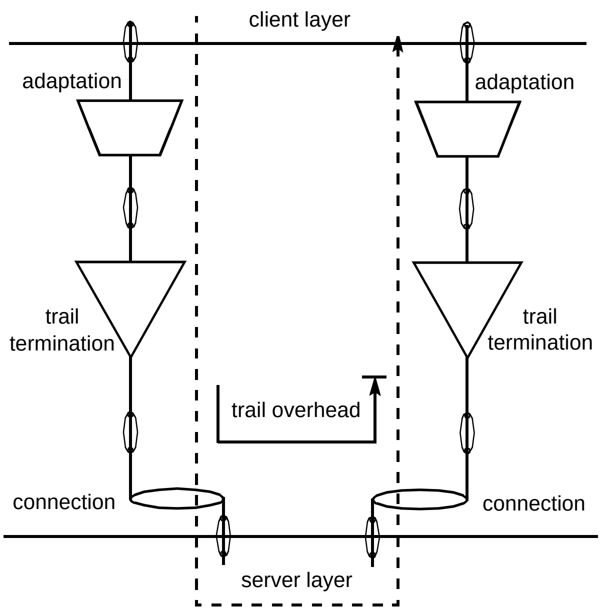

Fig 4 Functionality within a transport layer.

In the SDH environment, the data to be transported is mapped into a container. This mapping process accommodates any frequency deviation on the client data. The SDH path overhead is added to the container to monitor error performance, payload composition and path routeing. The overhead remains with the client data until it is mapped out of the associated network layer. The addition of a path overhead to a container forms an SDH virtual container (VC). The hierarchy defines virtual containers at four levels (VC-4, VC-3, VC-2 and VC-1). The way in which 2 Mbit/s constant bit rate data is mapped into the hierarchy is illustrated in Fig 5 [1].

In a PDH system, frequency differences between different signals are accommodated through bit justification. The synchronous digital hierarchy is based on byte structures and frequency differences between network nodes are accommodated through byte justifications. This justification process is controlled by a pointer mechanism. The addition of a pointer to a virtual container forms either a tributary unit (TU) or an administrative unit (AU) depending upon the level of the hierarchy in question. A pointer exists for every individual payload which can therefore be identified within a higher rate aggregate without the need for demultiplexing.

The logical relationship between a VC-4 and its parent STM-N is expressed in terms of a phase alignment of the VC-4 with its associated AU-4.

One of the most important considerations for any replacement scheme is that it must accommodate that which came before. The way in which this is achieved using the containers, virtual containers and administrative units currently defined for the synchronous digital hierarchy is illustrated in Fig 6. The mapping of a 6 Mbit/s signal into a VC-2, and the use of an AU-3 are not supported in Europe.

Figure 6 illustrates the way in which existing data services may be carried over SDH networks. There are several advantages of SDH which essentially address the disadvantages of PDH networks. They may be summarised as:

- SDH is the latest generation of transmission equipment, fully specified for operation up to 9.95 Gbit/s,
- direct access to tributary signals (add-drop) removes the multiplex mountain and hence improves the reliability,
- SDH equipment has a standardised optical interface which allows interconnection of equipment from different suppliers — there can, however, be specific difficulties with ‘SDH’ interfaces on CPE,
- SDH equipment and networks were designed for full remote software configuration and management,

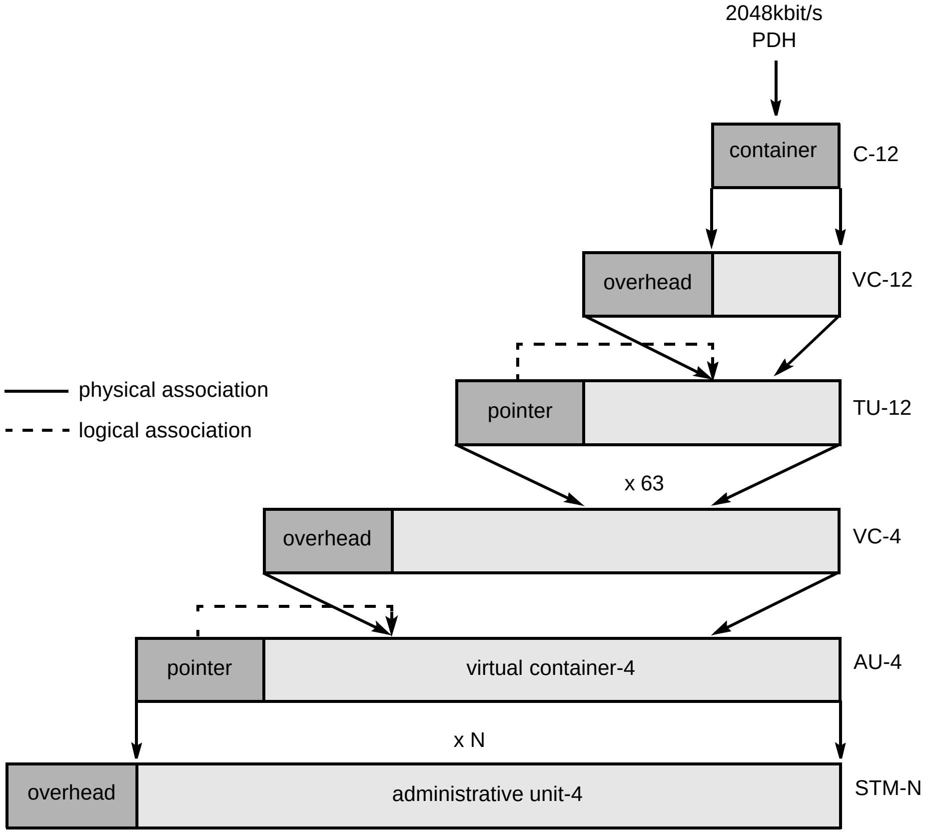

Fig 5 SDH mapping principles.

- SDH frame structure includes comprehensive overhead bytes for improved OAM and PM facilities,
- new network elements (ADM, crossconnects) enable the deployment of new network structures, e.g. self-healing rings.

The last point illustrates the difference in philosophy between PDH and SDH networks. PDH transmission systems were essentially point-to-point in nature whereas SDH systems have been designed for point-to-point, chain or ring applications. The benefits are best illustrated by

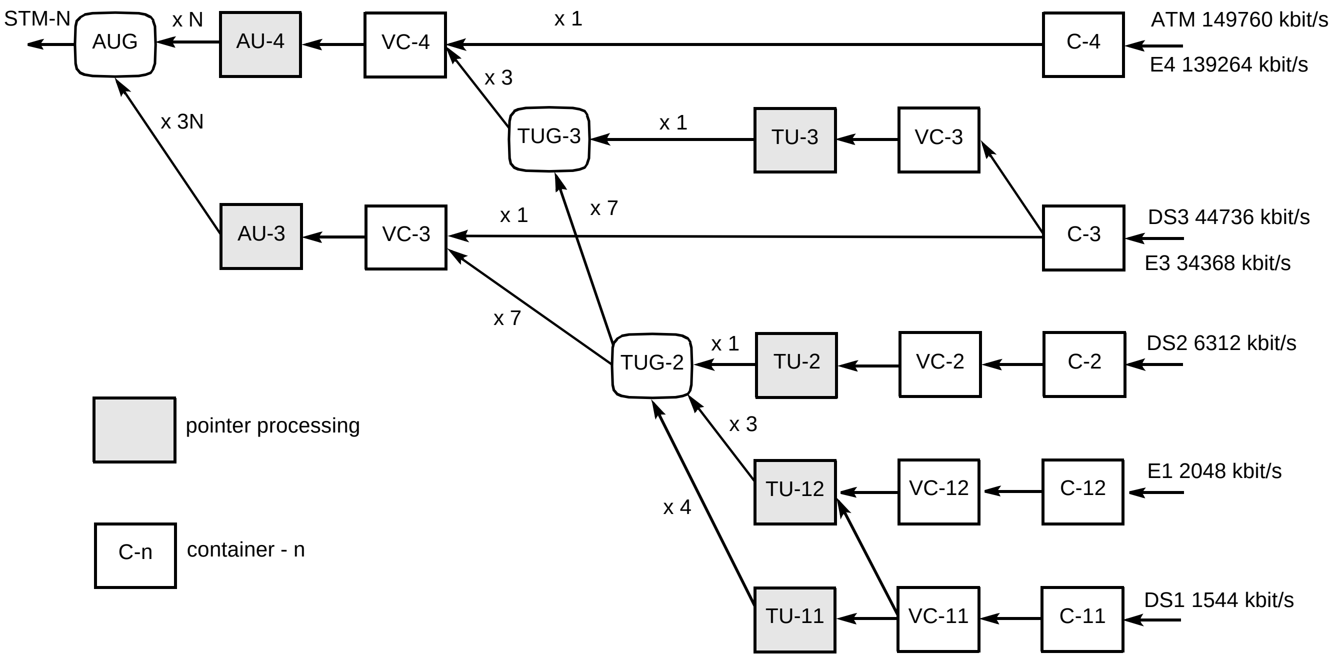

Fig 6 SDH mappings.

comparison between the use of a PDH system and an SDH chain as shown in Fig 7.

This illustrates the use of PDH and SDH systems to link two 64 kbit/s switches and shows the reduction in equipment volume achieved using the latter technique.

Recent experience has shown that customer demand for SDH services has to be tempered by the ability of the technology to deliver the service required by the customer. The most fundamental of the characteristics is timing transparency which is discussed elsewhere [7].

SDH networks also offer fast protection of services and the effects of this on the transported service must be appreciated. BT currently offers a fully resilient private circuit product called 'MegaStream Genus'. This offers end-to-end path duplication and service protection within 200 ms (200 ms is the product specification, 50 ms the equipment specification and 10-20 ms is the typical value for equipment).

The service architecture is illustrated in Fig 8.

The SDH equipment is specified to switch within 50 ms of the detection of a fault. This time was originally determined by the effect on PSTN services which tend to be intolerant of delay but tolerant of short breaks. The tolerance to short breaks and variations in path delay of data services is quite different and is discussed in the following section.

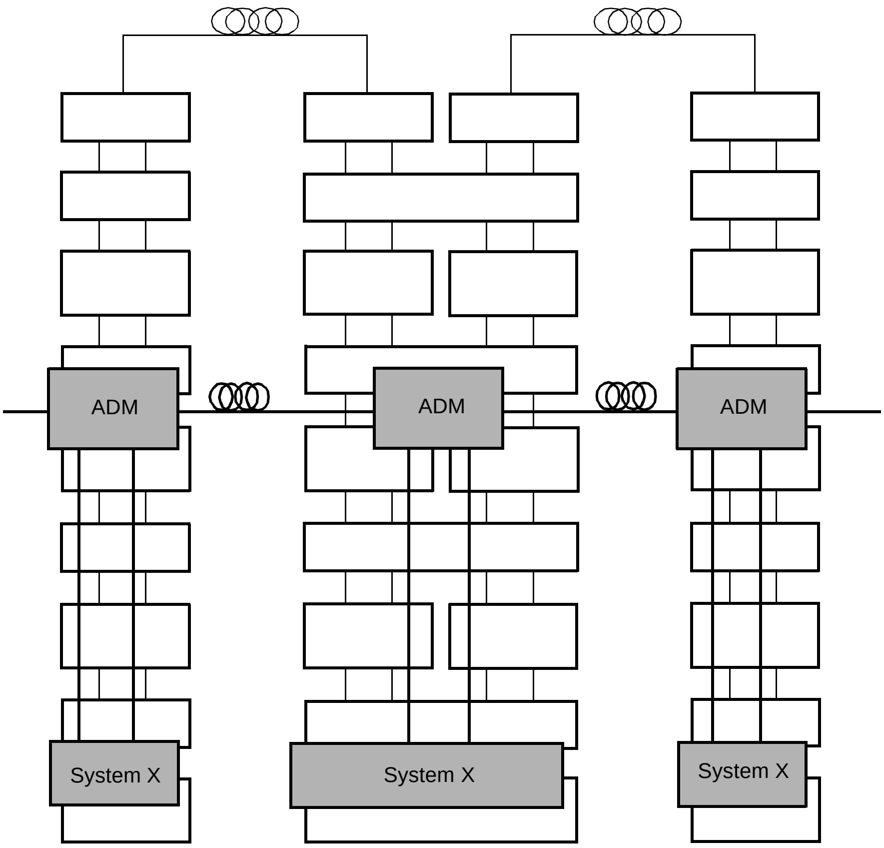

Fig 7 SDH multiplexer chain.

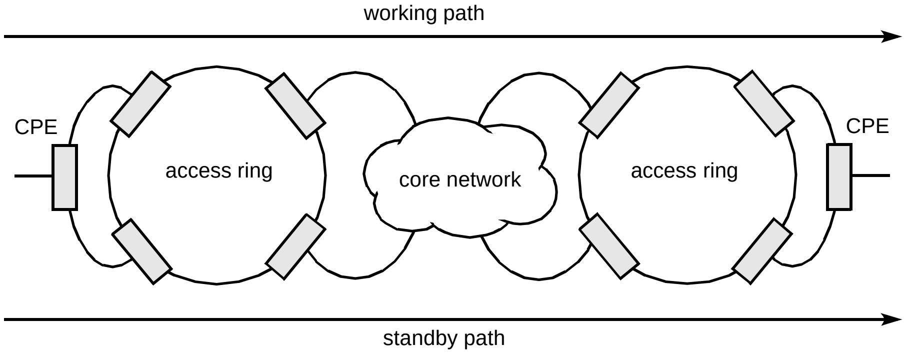

Fig 8 SDH private circuit delivery.

The synchronous digital hierarchy fully supports existing data services using PDH interfaces, but it also specifically supports new services as described below.

5.1 ATM over SDH

ATM cells may be mapped into the SDH frame at a variety of rates as is the case with the BT CellStream service [8]. At present, the most common rate is the VC-4 level although concatenated VC-4 payloads can be used to transport higher ATM bit rates. The mapping arrangements are illustrated in Fig 9 [1].

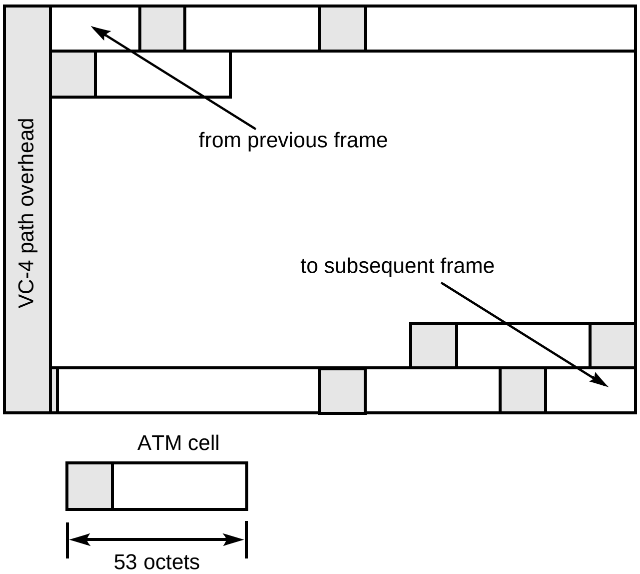

Fig 9 Mapping of ATM cells into a VC-4.

The mapping of ATM cells is performed by aligning the byte structure of every cell with the byte structure of the container. A virtual container is formed through the addition of a specific path overhead. The VC-4 container capacity is not an integer multiple of the ATM cell length (53 bytes) and so a cell may cross the container frame boundary.

Existing ATM services may be supported over the BT SDH network, but the transport of payloads in excess of 155 Mbit/s requires enhancement of the current generation of SDH equipment being deployed by BT. This is because ATM services are generally defined for transmission over SONET networks using contiguous concatenation of VC-4 payloads whereas the first generation of SDH equipment may only support virtual concatenation. The latter technique requires careful path management to avoid significant variations in path length.

ATM cells are subject to many processes as part of their transport across an SDH network. The cell stream is scrambled prior to insertion into the SDH container, the SDH signal is itself scrambled prior to transmission, the transmission process introduces degradation, and the reverse processes are applied before the cells are returned to the ATM layer for further processing. The degradation introduced by the transport network includes:

- bit errors, both single and burst, leading to cell corruption,
- timing jitter and wander caused by the mapping process,
- timing jitter and wander caused by the SDH pointer process.

The SDH equipment may protect a payload within 50 ms. This introduces a short break into the transported ATM VC/VP, and may alter the path length and hence the cell delay. This must be contrasted with the PDH network where slower 'restoration' tends to be used. Transport over PDH therefore requires rerouteing within the ATM layer whereas transport over SDH networks may not.

5.2 IP over SDH

IP packets may be transported over SDH networks as a PDH payload, but discussions to define a direct mapping of packets into an SDH VC are also currently taking place.

The point-to-point protocol (PPP) [9] provides a standard method for transporting multi-protocol datagrams over point-to-point links. Initial deployment has been over short local lines, leased lines, and POTS, using modems. As new packet services and higher speed lines are introduced, PPP may be easily deployed in these environments as well. The PPP encapsulation for high-speed point-to-point SONET/SDH links, which has been defined in RFC 1619 [10], has a relatively low overhead. It is anticipated by the IETF that significantly higher throughput can be attained compared to other SONET/SDH payload mappings, at a significantly lower cost for line termination equipment.

The basic rate for PPP transmission over SDH is that of STM-1 at 155.52 Mbit/s, with the available information bandwidth being 149.76 Mbit/s (SDH VC-4).This is the same super-rate mapping that is used for ATM and FDDI [10]. Lower signal rates use the virtual tributary mechanism of SONET (SDH lower order VC). This maps existing signals up to T3/E3 rates asynchronously or uses available clocks for bit-synchronous and byte-synchronous mapping. The IETF has recommended that higher signal rates should conform to the SDH STM series, rather than the SONET STS series. This is because the STM series progresses in powers of 4 (instead of 3), and employs fewer steps, which is likely to simplify multiplexing and integration.

PPP presents an octet interface to the physical layer. The framing for octet-synchronous links is described in Simpson [8]. PPP treats SDH transport as a full-duplex, octet-oriented synchronous link. There is no provision for suboctets to be supplied or accepted. The octet stream is mapped into the SDH virtual container with the octet boundaries aligned with the VC octet boundaries. The PPP variable length frames are located by row within the VC payload and allowed to cross VC boundaries where necessary. A 32-bit frame-check sequence (FCS) is recommended.

Path signal label (C2 byte) is intended to indicate the contents of the VC-4. A value of 207 (cf 'hex') has been proposed to indicate PPP. The SDH multi-frame indicator (H4) is currently not used. PPP does not require the use of control signals; when available, however, using such signals can allow greater functionality and performance. Implications are discussed in Simpson [11]. At such high speeds, address, control and protocol field compression options within PPP are not recommended.

Scrambling is presently not defined in RFC 1619 [11] for the insertion of PPP HDLC frames into the VC-4. The consequence of this is that there is the possibility of a malicious user repeatedly sending a 127-byte 'magic' sequence in the HDLC frames. This magic sequence, when aligned with the 127-byte SDH frame scrambler could result in the loss of SDH timing information due to consecutive zeros. The IETF and the ITU are presently discussing solutions to this issue with the use of self-scrambling [7] appearing to be the current favourite.

For backbones built expressly for providing Internet or IP data services, a routeing-based architecture using high bandwidth utilisation packet interfaces (packet over SONET) is usually most appropriate. This is because packet-based OC-3 networking equipment can match the interface bandwidths supported by asynchronous transfer mode with much lower overheads. Packet data is transported using point-to-point protocol which is mapped into the SDH STM-1 payload. These encapsulations accumulate 7 bytes of transport overhead per frame, which is less than half of that involved with ATM using ATM adaptation layer 5 (AAL5) and line card control (LCC) sub-network access protocol (SNAP) encapsulations. The use of PPP over SONET/SDH therefore maximises data throughput and line utilisation which is critical for expensive, high-bandwidth facilities. It should be noted that the encapsulation overheads are worse when carrying PPP over ATM over SDH as is sometimes proposed. To date, several commercial implementations of SDH interfaces on IP routers have been announced, most notably by Cisco.

The effects of transport layer degradation upon IP packets depend on whether SDH or PDH transmission is employed. A break on a PDH link requires IP layer restoration of the service. Traffic is interrupted for the duration of the re-route which depends on the IP routeing protocol deployed. Any short break on a protected SDH link results in SDH rapidly reconfiguring to restore the link before the loss is detected by the IP routeing protocol. In both cases, there is some loss of data packets and in many cases re-transmission which would be buffered at a higher layer (TCP) thereby maintaining the application.

Any change in path delay caused by the SDH protection operation is accommodated by buffering, as is the jitter and wander introduced by the link.

Burst or bit errors introduced by the transmission system may cause packet loss and re-transmission.

5.3 SMDS over SDH

The BT switched multi-megabit data service is implemented using MAN switches linked by either PDH or SDH transmission [13].

The long-distance trunks between switches are 'looped-buses' which in the event of a break on a PDH link would re-heal the ring after ~10 s. An outage that causes both routes round the ring to fail requires a full re-route, which can take several minutes. Traffic is interrupted for the duration of the re-heal or re-route.

Any short break on an SDH link is ignored although there is a loss of some data packets.

Any change in path delay caused by the SDH protection operation is ignored as is the jitter and wander introduced by the link.

Burst or bit errors introduced by the transmission system cause packet loss as the CRC checks fail and packets are discarded on re-assembly.

6. The effects of new bandwidth demands on transmission networks

To date, this paper has only considered the impact of technology upon the transport of data services, but one must also consider the impact of data services on technology.

The BT core SDH network in the UK was originally designed and dimensioned to serve a specific market although it has subsequently grown to provide transport for a number of data services. The explosion in demand for Internet services has meant that a networking rate of 2 Mbit/s is insufficient. If 2 — 20 Mbit/s becomes the accepted norm for home access, businesses may require 100 — 600 Mbit/s in the relatively short-term.

There are two obvious effects of this. The first is the effect on the existing revenue streams, but there is also the sheer infrastructure capacity required in order to transport the data. Significant growth in ATM, Internet protocol and other data services could see the core network transport requirements increase fivefold over the next few years. A growth in capacity and downward pressure on revenue are the twin challenges to be faced by those implementing broadband transport networks for BT.

7. Network deployment

BT, in common with other operators, continues to deploy expanding SDH networks. In the UK, the main core network is based on equipment and management systems [14] supplied by GPT as illustrated in Fig 10. However, BT also has a network of Ericsson SDH equipment to provide international backhaul capacity within the UK, is deploying SDH networks overseas in collaboration with our alliance partners, and has significant interests in many SDH submarine cables world-wide.

The UK core network was originally deployed to serve the 2 Mbit/s private circuit market but has grown significantly to serve both higher bit rate private circuits, SMDS and the CellStream ATM service. This growth has been achieved organically which is important to avoid the fragmentation of the network which can be characteristic of some other deployment strategies. Non-organic network growth can cause problems with the transportation of timing-sensitive services such as video.

The UK network continues to evolve to address new requirements such as the transportation of concatenated

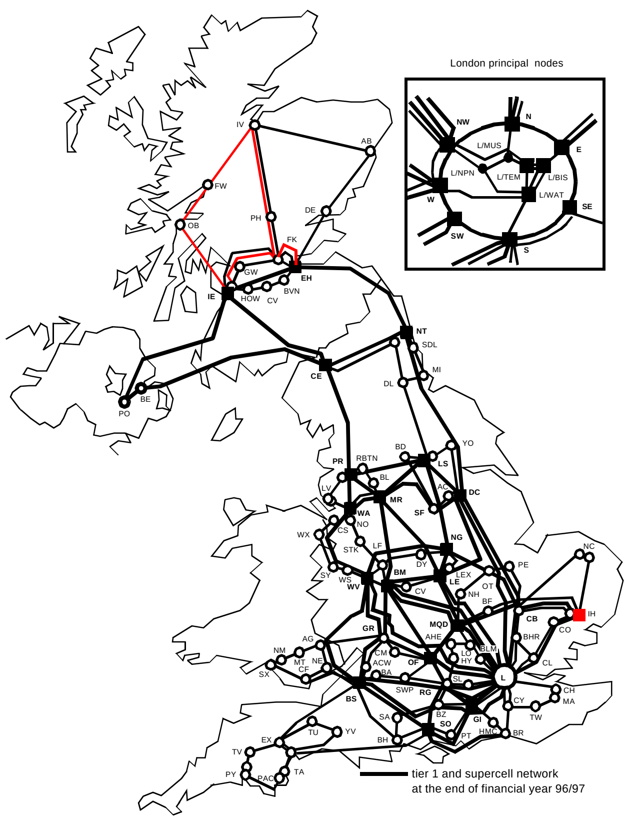

Fig 10 BT SDH network 1996/97.

VC-4s. This requires enhancements to the SDH pointer processor which were not envisaged during the original deployment. Techniques exist to allow the network to transport these payloads using virtual concatenation, but attempting to route, protect and monitor a number of separate VC-4s as a single payload will stretch the capabilities of any management system.

However, the BT core SDH network has demonstrated that it is able to meet the challenge of new requirements, and has proven to be the technology of choice for many large business customers. It is now faced with the challenge of providing broadband transport for huge volumes of data, a challenge which is being addressed through the deployment of techniques designed to exploit the capacity of the installed fibre base.

Wavelength division multiplexing is a technique which enables many optical channels to be carried on a single (pair of) optical fibre(s). The technique is analogous to the use of FM in the radio spectrum where many channels can be accommodated within the available bandwidth.

The principle is illustrated in Fig 11. Each information stream, which may be operating at 2.5 Gbit/s, is carried by a different wavelength or colour of light. The channels occupy different sections of the fibre transmission window and do not interfere with each other to any great extent. The sources, which may be either wavelength selected transmitters or transponders, which provide wavelength translation from other sources, are combined by a passive device for transmission on a single fibre. The channels may be amplified to overcome transmission losses and the system may operate over several hundred kilometres. At the destination, the channels are separated by a diffraction grating and delivered to their individual client systems.

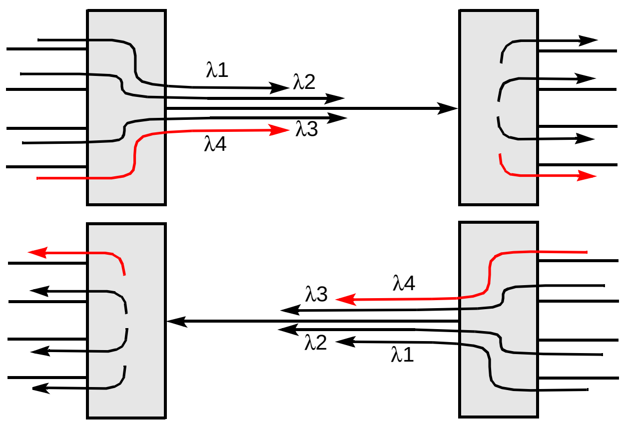

Fig 11 Point-to-point WDM system.

WDM systems may be used to provide direct connections between large network nodes or may just form a component of the overall transmission network.

The current generation of WDM systems may essentially be viewed as just another layer in the multiplexing hierarchy. Typically, they may combine a number of optical signals from SDH or PDH systems for transmission over a single fibre before the signal is again split for delivery to a number of clients. Certain WDM systems incorporate transponders which terminate the optical signal before re-transmitting the information on a predefined wavelength. This electro-optic process introduces a bandwidth constraint which may ultimately limit the bit rate of transmission. Other WDM systems rely on wavelength selection within the client systems to perform all-optical processing. The impact of these latter systems on the client transmission networks may simply be an increase in the optical loss of the fibre which can be accommodated during the system planning stage or indeed may be negated through the use of optical amplifiers. Such systems bring nearer the concept of a loss-less 'optical ether'.

As future systems may include routeing and protection within the optical domain, the possibility of disconnection or misrouteing within the optical layer must be considered. Furthermore, the use of optical amplifiers enables an increase in system span length which, for high bit rate systems, introduces nonlinear effects previously not considered for terrestrial systems.

8. Conclusions

Modern transport networks must accommodate a variety of client services.

The PDH systems which have served operators for a number of years are gradually being replaced by SDH networks designed to offer improved performance and flexibility.

Each of these technologies introduces different types of degradation of the transported signals; specifically, the timing transfer characteristics of SDH and PDH systems are different. SDH systems cannot be characterised as simply 'managed PDH' without consideration of the service being carried.

Acknowledgements

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References

1 ITU-T Recommendation G.707: 'Network Node Interface For The Synchronous Digital Hierarchy (SDH)', (June 1996).

2 ANSI: 'American National Standard for Telecommunications — Synchronous Optical Network (SONET) Payload Mappings', ANSI T1.105.02-1993 (1993).

3 ITU-T Recommendation G.803: 'Transport Network Architecture', (1993).

4 ETS 300 417-1: 'Generic Functional Requirements for SDH Equipment', (1995).

5 ITU-T Recommendation G.702: 'Digital Hierarchy Bit Rates', (1993).

6 ITU-T Recommendation G.703: 'Physical/Electrical Characteristics of Hierarchical Digital Interfaces', (1991).

7 Ferguson D et al: 'Self-Synchronous Scramblers for PPP over Sonet/SDH', (ftp:ietf.org/internet-drafts/draft-ferguson-pppsonet-selfsync-00.txt) (1997).

8 Chauhan J and Gallagher I D: 'CellStream — pilot customer experience', BT Technol J, 16, No 1, pp 88—95 (January 1998).

9 Simpson W (Ed): 'The Point-to-Point Protocol (PPP)', RFC 1548, Daydreamer (December 1993).

10 Simpson W (Ed): 'PPP in HDLC Framing', RFC 1662, Daydreamer, (July 1994).

11 Simpson W (Ed): 'PPP over SONET/SDH', RFC 1619 (May 1994).

12 Brown T S et al: 'Synchronisation in data networks', BT Technol J, 16, No 1, pp 159—170 (January 1998).

13 Lewis D J et al: 'Four years of the Switched Multi-megabit Data Service', BT Technol J, 16, No 1, pp 16—24 (January 1998).

14 Cole R V et al: 'The role of data networking in the management of BT's SDH network', BT Technol J, 16, No 1, pp 171—181 (January 1998).

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Dave Rowland joined BT in 1973 as an apprentice. After completing his apprenticeship he worked on the maintenance of BT's national and interrnational transmission network. In 1988 he moved to a transmission support role; during this period he influenced the development and deployment of transmission surveillance systems and contributed to a number of operational effectiveness measures.

In 1995 he joined the SDH technology team at BT Laboratories.

His current work supports BT's national and international SDH transmission networks and systems.

Alex Vinall joined BT in 1993 after graduating from the University of Manchester Institute of Science and Technology with an MEng degree in Microelectronic Systems Engineering. His initial work focused on the development of management system interfaces for SDH, in particular for international applications.

He joined the SDH technology team in 1994 and has played a significant role in the development and enhancement of BT's international back-haul SDH network. He has undertaken an ongoing programme of tests which ensure successful interworking between BT's SDH equipment and that used by other licensed operators.

Alan O'Neill joined BT as an apprentice technician in 1981. He was sponsored by BT to attend the University of Warwick from 1986 to 1989 and gained his BSc in Physical Electronics. On joining BT Laboratories, he spent three years investigating advanced switching, transmission and control architectures based on optical and opto-electronic techniques. He then spent two years on customer private ATM networks before helping CONCERT to design and implement the network testing of the joint BT-MCI voice network platform. He gained his BT Martlesham MSc from University College London in 1994 and since then has been developing a team of researchers in the area of Internet transport research. This is closely aligned to the work of the Internet Engineering Task Force and he is presently acting as the BT standards co-ordinator for this body. Convergence of IP/ATM and IP/IN continue to be significant study areas for the team as well as novel evolutionary scenarios for the elements of the Internet protocol suite.